

NODIRBEK MAKHTUMOV

AI/ML Engineer

Computer Vision · Edge AI · Model Optimization

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Daejeon, South Korea

E-7 Special Ability · Authorized to work in Korea

Summary

AI/ML engineer with 3+ years building and shipping production computer vision systems that run on edge hardware. I own the full path — model design and training, TensorRT/ONNX quantization, deployment to NVIDIA Jetson and Hailo NPU, FastAPI services and Docker-based edge operation. I validate models with per-class metrics, adversarial robustness testing and per-hardware measurement rather than a single headline accuracy figure.

Selected Results

30%	25+ FPS	8.6×	76% → 92%
GPU memory reduction	4 concurrent streams	NPU inference speedup	OCR accuracy gain

Technical Stack

Computer Vision	Object Detection, Semantic / Instance Segmentation, OCR, Multi-Object Tracking, Face Recognition (ArcFace)
Frameworks	PyTorch, TensorFlow, Keras, Ultralytics (YOLO), OpenCV, InsightFace, Hugging Face Transformers
Model Optimization	Quantization, Pruning, TensorRT, ONNX Runtime, DLA offloading, backbone replacement and A/B evaluation
Edge Hardware	NVIDIA Jetson (Orin NX · Orin Nano), Hailo-8 NPU, Raspberry Pi 4/5, Android tablets
Deployment	Docker, FastAPI, Flask, REST API, PostgreSQL, concurrent multi-stream serving
Evaluation	Per-class metrics, adversarial robustness testing, Grad-CAM, FPS/latency/jitter benchmarking, W&B, MLflow
Languages & Tools	Python (primary), C++, SQL, Linux (Ubuntu), Git/GitHub, GitLab, GitHub Actions CI

Experience

Netvision Telecom AI/ML Engineer · Software Development Team 2023.04 – Present

Joined as a paid research intern in April 2023, converted to full-time

- Smart waste collection fleet management** — Built YOLO-based waste classification and volume estimation. Replaced the YOLOv8 backbone with ResNet50 using 1x1 conv channel adapters to match the PANet neck, offloading ResNet layers to the Jetson DLA for a **~30% GPU memory reduction**. Deployed to vehicle-mounted edge inference after quantization and TensorRT conversion
- Edge-aware segmentation pipeline (rail-wheel gap measurement)** — Combined a lightweight segmentation model with RANSAC boundary extraction and a custom point-stabilization algorithm to correct curved boundaries. FastAPI serving plus PostgreSQL metadata logging, sustaining **25+ FPS across 4 concurrent camera streams**
- SAM2 automated video labeling tool** — Built a pipeline that propagates labels across a full video from a single first-frame prompt and exports to YOLO and COCO formats, giving a **10x+ speedup** in dataset construction
- (Research intern, 2023.04–2023.09)** Built and deployed YOLO vision models on Jetson Orin Nano, applied latency optimization and model quantization, collected and labeled training datasets

Experience (continued)

Freelance contract

AI/ML Engineer — Face recognition attendance kiosk

2026.07

- Implemented the same recognition pipeline three ways — **Python/CPU, a C++ port, and Hailo-8 NPU** — and benchmarked them on identical Pi 5 hardware: median latency 48.08ms → 5.59ms (**8.6×**), jitter ±27ms → ±0.29ms (**95×** tighter)
- SCRFD/RetinaFace detection with ArcFace 512-d embeddings (InsightFace); built an iPhone-Face-ID-style five-pose guided enrollment wizard
- Used the measurements to select **Samsung Galaxy Tab** as the deployment target and ported the pipeline to on-device inference for delivery. Offline-first SQLite (WAL) logging with Google Sheets sync

Freelance contract

AI/ML Engineer — Smart truck parking & vehicle ID recognition

2025.09 – 2025.11

- Built the truck detection, ID recognition and live occupancy streaming system **end-to-end, solo**
- Systematic error analysis identified perspective distortion as the dominant failure mode; built a **custom deskewing module** (contour detection + perspective transform) feeding EasyOCR — **ID recognition accuracy 76% → 92%**
- Streamed occupancy state and recognized IDs to the backend over a REST API and into the operations dashboard
- Published a clean open-source version with a rewritten deskew module — on synthetic tilted plates, EasyOCR accuracy **80.5% → 97.5%** (github.com/Nodir0705/smart-truck-parking)

Projects & Other Activities

IoT security gateway (Orange Pi 5 · RK3588S) · 2026 · github.com/Nodir0705/iot-gateway-security

Combined XDP/eBPF kernel filtering, 244 IoT-specific Suricata rules, a flow autoencoder on the RK3588 NPU and behavioral detectors (beaconing, DGA DNS), with a 60-second evidence score that triggers automatic nftables quarantine. In a 47-minute stress test it **caught 9/9 attack phases** and **87/88 (98.9%)** of expected rules fired

Adversarial robustness of a malware image classifier · Personal project · 2026 · github.com/Nodir0705/malware_classifier

Validated a MallImg ResNet50 (25 families, 99.48% clean accuracy) under PGD attack — changing each byte by at most 2 or 4 values dropped it to 22% and 6.6%. Full-network PGD adversarial retraining restored **accuracy under attack from 22% to 94.15%** for a 2.3-point clean-accuracy cost. Reproducible code and results published, with Grad-CAM interpretability

Mentor · Woosong University Capstone · 2024

Guided a computer vision team (image processing, YOLO training) with weekly feedback; contributed to a 1st-place finish among 200+ competing teams

Education

BSc, Computer Science

Woosong University · Daejeon

2019 – 2023

Languages

Korean	Conversational
English	Fluent — professional working proficiency
Russian	Conversational

Visa

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